

Exact K3P Tree–Theta-Trinet Collision

Technical summary

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Resolution. The high-level K3P trinet question has a negative answer: the K3P analogue of the high-level trinet inequality fails. A fixed binary semi-directed strict level-two theta trinet with a genuine nontrivial 3-blob produces exactly the same three-leaf distribution as an admissible three-star tree. The witness lies in the quartic number field $\mathbb{Q}(h)$, where $h = 5^{-1/4}$ is isolated by $2/3 < h < 7/10$; exact calculations use the basis $1, h, h^2, h^3$. Both inheritance probabilities are $1/2$, every nontrivial edge eigenvalue lies in $(0, 1)$, and every transition probability is strictly positive. This answers negatively the K3P high-level trinet question raised by Brits, Holtgreffe, van Iersel, and Martin after their level-one K3P result and arbitrary-level JC/K2P results.

Topology and root suppression. Use the rooted calculation network

$$\rho \rightarrow 1, \quad \rho \rightarrow u, \quad u \rightarrow p, q, \quad p, q \rightarrow r_2, \quad p, q \rightarrow r_3, \quad r_2 \rightarrow 2, \quad r_3 \rightarrow 3.$$

After suppressing ρ , the two root edges combine into the semi-directed pendant edge $1 - u$. The remaining blob is the theta graph with paths

$$p - u - q, \quad p - r_2 - q, \quad p - r_3 - q.$$

Its three incident cut edges lead to leaves $1, 2, 3$, and it contains exactly the two reticulations r_2, r_3 . Thus it is a strict level-two nontrivial 3-blob. No no-2-sub-blob convention is needed.

Exact network parameters. In coordinate order (A, C, G, T) , let

$$\begin{aligned} K &= (1, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}), & U &= (1, \tfrac{h}{3}, h, \tfrac{1}{3}), & V &= (1, h, \tfrac{h}{3}, \tfrac{1}{3}), \\ S &= (1, \tfrac{3h^2}{4}, \tfrac{1}{4}, \tfrac{3}{10}), & T &= (1, \tfrac{1}{4}, \tfrac{3h^2}{4}, \tfrac{3}{10}). \end{aligned}$$

Assign K to $\rho \rightarrow 1, \rho \rightarrow u, r_2 \rightarrow 2, r_3 \rightarrow 3$; U, V to $u \rightarrow p, u \rightarrow q$; and S, T to all incoming reticulation edges from p, q , respectively. Root suppression gives

$$K \odot K = (1, \tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}), \quad p(K \odot K) = (\tfrac{7}{16}, \tfrac{3}{16}, \tfrac{3}{16}, \tfrac{3}{16}).$$

All inverse Fourier edge probabilities are strictly positive. The explicit U, V satisfy

$$U_C = U_G U_T, \quad V_G = V_C V_T,$$

so the closed-form witness is on the boundary of the smaller strict positive-rate continuous-time cone, but strictly inside the stated stochastic space Θ_0 .

Comparison tree and factorization. Set

$$\alpha = (1, \tfrac{5h^3+h}{16}, \tfrac{5h^3+h}{16}, \tfrac{h^2}{4}), \quad \beta = \gamma = (1, \tfrac{h^2}{4}, \tfrac{h^2}{4}, \tfrac{h^2}{4}).$$

These three edges have strictly positive transition probabilities and strictly positive continuous-time K3P rates. For a consistent label (x, y, z) , write $x = y + z$. Removing the four K factors, the network core is

$$M_{y,z} = \tfrac{1}{4}(S_y S_z U_{y+z} + S_y T_z U_y V_z + T_y S_z U_z V_y + T_y T_z V_{y+z}).$$

Define

$$B = (1, \tfrac{h^2}{2}, \tfrac{h^2}{2}, \tfrac{h^2}{2}), \quad P = (1, \tfrac{5h^3+h}{4}, \tfrac{5h^3+h}{4}, h^2).$$

Exact reduction using only $5h^4 = 1$ gives

$$\boxed{M_{y,z} = P_{y+z} B_y B_z} \quad (y, z \in \mathbb{Z}_2^2).$$

Therefore

$$q_{xyz}^N = K_x^2 K_y K_z M_{y,z} = \alpha_x \beta_y \gamma_z = q_{xyz}^T.$$

There are sixteen consistent Fourier coordinates and forty-eight inconsistent coordinates, which vanish in both models. Hence all sixty-four Fourier coordinates and, by invertibility of the finite Fourier transform, all sixty-four leaf-pattern probabilities agree exactly.

Full-rank strengthening. Differentiate the fifteen nonconstant consistent Fourier coordinates in row order

$$\begin{aligned} ACC, AGG, ATT, CAC, CCA, CGT, CTG, GAG, \\ GCT, GGA, GTC, TAT, TCG, TGC, TTA \end{aligned}$$

with respect to the specified fifteen edge/mixing coordinates (three coordinates on $\rho \rightarrow 1$, a_G on $u \rightarrow p$, a_C, a_G on each incoming reticulation edge, a_T on each reticulation-to-leaf edge, and the p -parent weight at r_3). Exact elimination gives

$$\det J_* = \frac{h(10h^2 + 1)}{2^{61}3^45^{14}} > 0.$$

Thus the fixed theta-trinet parameter map has rank 15, so its stochastic image contains an ordinary open neighborhood of the common distribution in the 15-dimensional affine space of consistent three-leaf group-based Fourier coordinates, $q_{AAA} = 1$. Intersecting this ambient neighborhood with the tree model yields a relatively open tree-model neighborhood of collisions. This is a statement about distribution-space openness, not a claim that generic network parameters are tree-equivalent.

Strict continuous-time strengthening. For a continuous-time K3P edge with all three rates positive, the exact equivalent conditions are

$$a_C > a_G a_T, \quad a_G > a_C a_T, \quad a_T > a_C a_G.$$

Increase the two saturated coordinates by a common $\varepsilon > 0$:

$$U_C \mapsto U_C + \varepsilon, \quad V_G \mapsto V_G + \varepsilon,$$

and solve for the fifteen pivot parameters while holding the fifteen nonconstant Fourier coordinates fixed. A real-analytic implicit-function argument gives a branch satisfying the fixed-output identity exactly for all sufficiently small ε . Its exactly verified tangent satisfies $J_* p' + F_{U_C} + F_{V_G} = 0$. In particular,

$$\frac{da_G^U}{d\varepsilon} = \frac{6}{1 + 10h^2},$$

and the two formerly saturated margins have derivatives

$$\left. \frac{d}{d\varepsilon}(U_C - U_G U_T) \right|_0 = \frac{10h^2 - 1}{1 + 10h^2} > 0, \quad \left. \frac{d}{d\varepsilon}(V_G - V_C V_T) \right|_0 = 1.$$

Every other rate margin, stochastic inequality, and mixing inequality has strict slack at $\varepsilon = 0$, so all persist for sufficiently small positive ε . Hence the same fixed tree distribution has a nearby theta-network preimage in which every network edge has three strictly positive continuous-time K3P rates.

What the result does and does not show.

- It shows that the K3P analogue of the high-level trinet inequality fails and blocks the direct K3P extension of the JC/K2P global blob-detection corollary.
- It shows that a nontrivial 3-blob is not universally identifiable from three-leaf K3P distributions on Θ_0 .
- It does not settle K3P quartet distinguishability, classify all tree/theta intersections, or imply generic tree-equivalence of theta-network parameters.

Exact audit. The machine-readable certificate stores the graph, all edge probabilities, all sixteen core entries, all sixty-four Fourier coordinates and pattern probabilities, the full Jacobian minor, and the exact tangent identity and strict-margin signs used in the real-analytic implicit-function argument. From the package directory run

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python3 verify.py
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using only the Python standard library. Successful output ends with

ALL EXACT CHECKS PASSED.